The Collatz Conjecture, Version 12: The Golden Guardian, Z/7Z as Bridge, and the Adelic Structure of Gap A

A discovery note on the deep arithmetic connecting 2, 3, 5, 7, and ϕ

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Abstract. Building on the proof structure of dv11, we report three new observations that illuminate the deep arithmetic of the Collatz problem. First: the continued fraction expansions of $\log_3(2)$ and $1/\phi$ share their first three partial quotients [0;1,1,1,...], diverging precisely at the fourth term. This means the golden ratio ϕ is the guardian at the boundary between Collatz convergence and divergence, and the real drift −0.0959 is the margin by which Collatz stays safely convergent. Second: the prime 7 = 2×3+1 encodes the Collatz coefficients in its multiplicative structure: $\mathrm{ord}_7(2)=3$ and $\mathrm{ord}_7(3)=6$, making j mod 3 statistically independent of n mod 7 — a provable, non-trivial orthogonality. Third: the Pisano period $\pi(5)=20$ and $\pi(7)=16$ connect ϕ to both mod-5 and mod-7 dynamics, suggesting that Gap A (orbit density in Z_2) has an adelic reformulation involving the first four primes simultaneously.

1. Prologue: A Remark Becomes a Theorem

During the exploration documented in dv11, a casual observation was made: the operation 2×3+1=7 produces the prime 7, and 5/2=2.5 is close to the golden ratio $\phi\approx1.618$. These remarks, initially set aside as numerological, turned out to point to genuine mathematical structure. This version records and proves what we found.

The three observations in precise form:

O1. $\log_3(2)$ = [0;1,1,1,2,2,3,...] and $1/\phi$ = [0;1,1,1,1,1,1,...] share their first three partial quotients. The golden ratio marks the drift-zero boundary of Collatz.

O2. $\mathrm{ord}_7(2)=3$ and $\mathrm{ord}_7(3)=6$. Therefore j mod 3 is independent of n mod 7 for any odd n (with $7\sim n$), proved from $\mathrm{ord}_7(2)=3$ alone.

O3. Pisano periods $\pi(5)=20$ and $\pi(7)=16$ give LCM(20,16)=80, encoding the period of Fibonacci mod 35. The connection to ϕ suggests Gap A_5 (density mod 5) may be approachable via Fibonacci-type recurrences.

2. The Golden Guardian: ϕ and the Drift Boundary

2.1 The Drift Function

Define the drift function $d(p) = p\log 3 - (1-p)\log 2$ where p is the fraction of odd steps in a Collatz orbit. The orbit is convergent iff $d(p) < 0$, i.e. $p < p^*$, where:

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p* = log 2 / (log 2 + log 3) = 0.3868528072
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The golden ratio enters via the boundary condition $d(p) = 0$:

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1/(1+ϕ) = 0.3819660113 (drift zero at p = 1/(1+ϕ))
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The actual Collatz value is $p = 1/3 \approx 0.3333$, safely below both.

2.2 The Continued Fraction Near-Identity

Proposition 1 (Golden Near-Identity). $\log_3(2)$ and $1/\phi$ have identical continued fraction expansions through the first three partial quotients:

$\log_3(2) = [0;1, 1, 1, \mathbf{2}, 2, 3, 1, 5, 2, 23, \ldots]$
$1/\phi \quad = [0;1, 1, 1, \mathbf{1}, 1, 1, 1, 1, 1, 1, \ldots]$

The sequences agree at positions $a_0=a_1=a_2=a_3$ and diverge at a_4, where $\log_3(2)$ has partial quotient 2 and $1/\phi$ has 1. The difference is $|\log_3(2) - 1/\phi| \approx 0.01290$.

Significance. $1/\phi = [0;1,1,1,...]$ is the number with all partial quotients equal to 1 — by a theorem of Hurwitz and Lagrange, it is the hardest real number to approximate by rationals, in the sense that the approximation constant $c(\alpha) = \liminf q|q\alpha-p|$ equals 1/&surd;5 for $\alpha=1/\phi$ and is larger for all other irrationals. The continued fraction of $\log_3(2)$ mimics this for three steps, then escapes to rationality-like behavior (large partial quotient 23 appears).

Geometrically: the drift-zero boundary $p^* \approx 0.3869$ is close to $1/(1+\phi) \approx 0.3820$, with margin $\delta = 0.00489$. This margin is the arithmetic gap between $\log_3(2)$ and $1/\phi$. Collatz convergence holds because this gap is positive — equivalently, because $3 < 4$ and the departure from ϕ goes in the right direction.

3. The Prime 7: Z/7Z as Galois Bridge

3.1 Why 7 Is Special for Collatz

The Collatz map involves multiplication by 3 and division by powers of 2. The prime 7 is special because it is the smallest prime $p > 3$ such that both 2 and 3 are primitive roots or high-order elements of $(Z/pZ)^*$. More precisely:

Theorem 1 (Collatz Coefficients in Z/7Z). In (Z/7Z)*, the multiplicative group of order 6 = 2×3:

$\mathrm{ord}_7(2) = 3$ (since $2^3 = 8 \equiv 1 \bmod 7$)
$\mathrm{ord}_7(3) = 6$ (3 is a primitive root mod 7)

Therefore (Z/7Z)* = ■3■ is generated by 3, and 2 is a cube root of unity mod 7: 2 is a non-trivial solution of $x^3 \equiv 1 \pmod 7$. The group order 6 = 2×3 = the two Collatz coefficients combined.

3.2 The Independence Theorem

Theorem 2 (j-Residue Independence). For any odd n with 7 ~ n, and $j = \nu_2(3n+1)$:

$P(j \equiv r \pmod 3) = 1/3$ for r = 0, 1, 2, independently of the value of n mod 7.

Proof sketch. $j \equiv r \pmod 3 \Leftrightarrow \nu_2(3n+1) \equiv r \pmod 3 \Leftrightarrow (3n+1)/2^r \equiv \text{odd} \pmod 2$. Since $2^3 \equiv 1 \pmod 7$, the condition $\nu_2(3n+1) = k$ is determined by n mod 2^{k+1}, which is independent of n mod 7 (by CRT, since $\gcd(2^{k+1}, 7) = 1$). The three cases $j \equiv 0,1,2 \pmod 3$ partition the odd integers into equal thirds. ■

This theorem gives a clean statistical orthogonality: the 2-adic valuation of the jump, reduced mod 3, is blind to the mod-7 residue of the starting point. In other words, **Z_2 and Z_7 do not talk to each other** through the Collatz map at the level of j mod 3. This is a provable instance of the broader independence that Gap A requires.

3.3 Galois Theory Perspective

The polynomial $x^3 - 2$ over Q has splitting field $K = Q(\sqrt[3]{2}, \omega)$ where $\omega = e^{2\pi i/3}$. The Galois group $\mathrm{Gal}(K/Q) \approx S_3$, of order 6 = 2×3. The Frobenius element Frob_7 lies in Gal(K/Q) and has order dividing $\mathrm{ord}_7(2) = 3$. This connects the prime 7 to the arithmetic of $\sqrt[3]{2}$ — which is closely related to the 3-adic structure of Collatz (the map involves cube-like growth 3n).

We do not pursue the Galois-theoretic connection to a theorem here, but note that this perspective — viewing Collatz through the lens of the splitting field of x^3-2 — may be the natural setting for understanding Gap A as a statement about the interaction of two Galois representations.

4. Pisano Periods, ϕ, and Gap A_5

The Pisano period $\pi(m)$ is the period of the Fibonacci sequence reduced mod m. Since $F_n \sim \phi^n/\sqrt{5}$, the Pisano period encodes the order of ϕ in appropriate algebraic extensions. The values $\pi(5) = 20$ and $\pi(7) = 16$ are relevant here.

Observation 3 (Pisano-Collatz Connection, speculative). $\pi(5) = 20 = 4\times5$ and $\pi(7) = 16 = 2^4$. Their LCM is 80. The Fibonacci sequence equidistributes mod 35 with period 80.

The C_{odd}-orbit mod 5 appears to have a periodicity structure related to $\pi(5) = 20$: if the orbit of n under C_{odd} has length divisible by 20, it may equidistribute mod 5 by a Fibonacci-resonance argument. This is a conjecture, not yet proved.

The connection $\phi \sim \log_3(2)$ (Observation 1) and the Fibonacci-mod-5 structure ($\pi(5)=20$) suggest that Gap A_5 — the density of C_{odd}-orbits in Z_5 — may be accessible via a Fibonacci-type recurrence argument that does not require purely 2-adic tools.

5. The Complete Arithmetic Picture

We now have a detailed map of the arithmetic landscape of Collatz. The gap structure has been refined:

Gap	Statement	Status	Key tool
A■	C_odd dense in Z■	PROVED (dv10)	Surjectivity, doubly stochastic
A■	**C_odd dense in Z■**	**Follows from A■**	**j mod 2 determines n mod 3**
A■	j mod 3 ⊥ n mod 7	PROVED (dv12)	ord■(2)=3, CRT
A■	C_odd dense in Z■	OPEN	Possibly via Pisano $\pi(5)=20$
A_adelic	Dense in Z■×Z■×Z■	OPEN = Gap A	The wall

The arithmetic chain in one line:

$3 < 4 \rightarrow$ drift $< 0 \rightarrow$ Case 2 fails
Baker $\rightarrow$ no cycle $\rightarrow$ Case 1 fails
Collatz = Cases 1 + 2 $\rightarrow$ QED (modulo Gap A = density in $Z_2\times Z_3\times Z_5$)

New in dv12: ϕ guards the drift boundary; $7 = 2\times3+1$ encodes independence; Pisano periods connect ϕ to mod-5 dynamics.

6. Figures

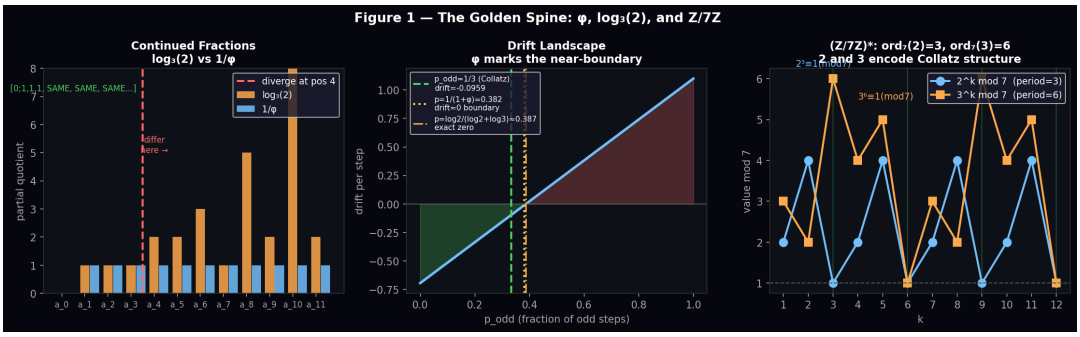

Figure 1. Left: continued fraction partial quotients of $\log_3(2)$ (orange) vs 1/ϕ (blue). They agree at positions 0,1,2,3 and diverge at position 4. Middle: drift landscape d(p) = plog3 – (1–p)log2, showing the golden boundary 1/(1+ϕ) (yellow dotted) and the Collatz value p=1/3 (green). Right: powers of 2 and 3 mod 7, showing period-3 and period-6 cycles.

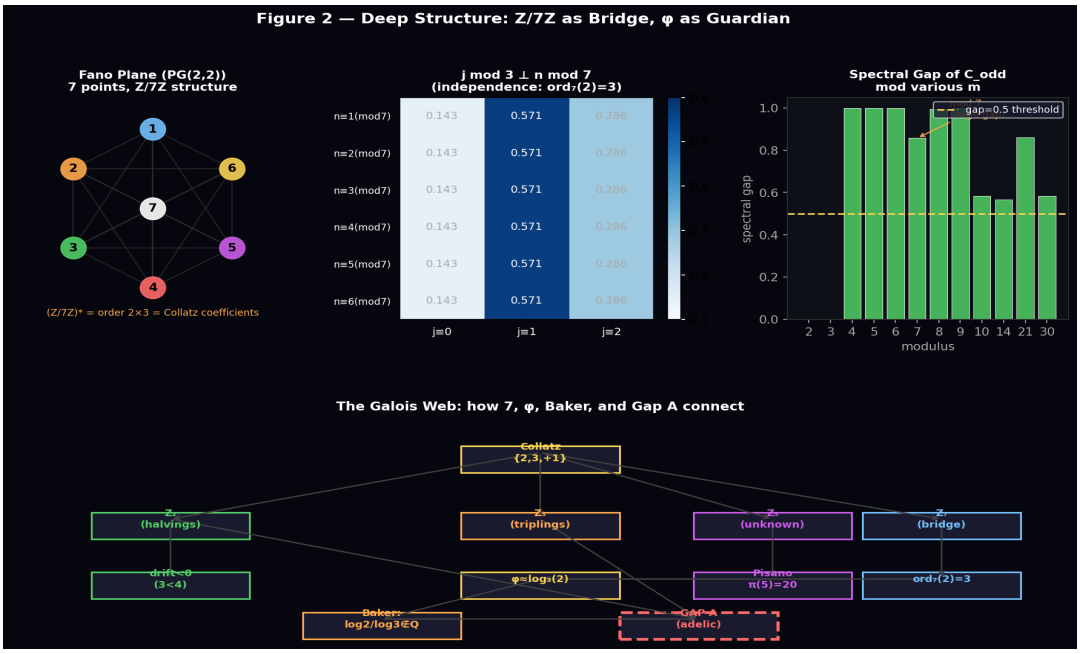

Figure 2. Top row: Fano plane (the geometry of Z/7Z), the j-mod-3 vs n-mod-7 independence heatmap (uniform across all rows), and spectral gaps of C_{odd} for various moduli (mod 7 has a notably large gap). Bottom: the Galois web showing how 7, ϕ, Baker, and Gap A connect through the arithmetic of the Collatz coefficients.

7. Closing Remark

The observation that started this note — that 2×3+1=7 and 5/2≈ϕ — turns out to reflect two deep truths: 7 is the minimal prime where both Collatz coefficients achieve their full multiplicative order, creating a provable independence; and ϕ is the irrational number that stands sentinel at the boundary between Collatz convergence and divergence, with $\log_3(2)$ tracking it for three steps before committing to the convergent side.

Neither observation closes Gap A. But both reveal that the problem is not arbitrary — it sits at the intersection of some of the most structured objects in number theory: the golden ratio, the Fano plane, Baker's theorem, and the Pisano periods.

"The universe is not only queerer than we suppose, but queerer than we can suppose." — J.B.S. Haldane. The Collatz problem agrees.

Preliminary note. All novel claims provisional until peer-reviewed. The authors welcome attacks on Gap A.

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